

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

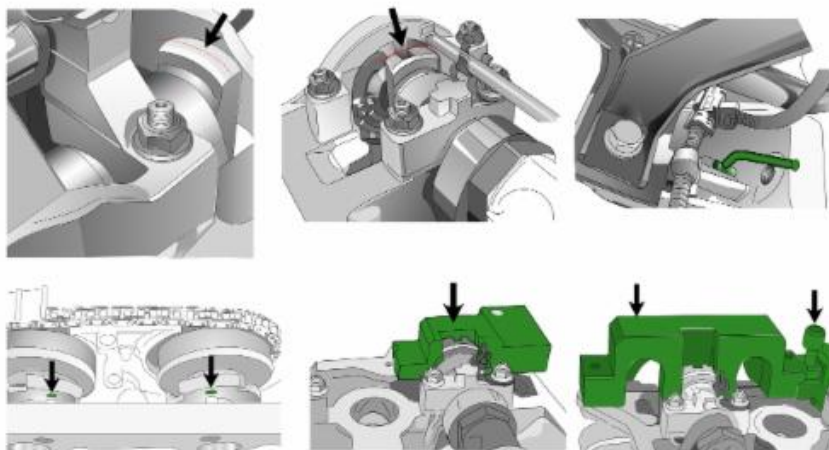
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

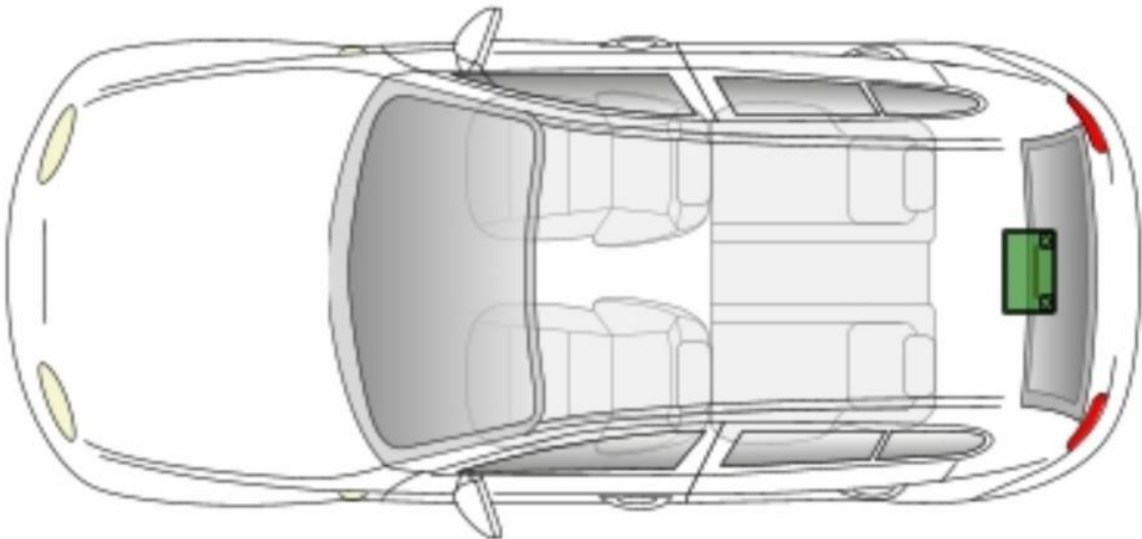
General

Before disconnecting the battery cable, check the audio system security code



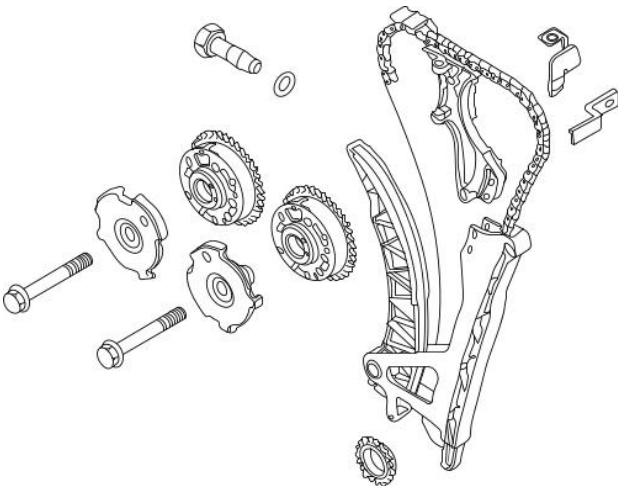
Battery location

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Luggage compartment

Timing chain layout



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Removal

Disconnect the battery earth cable

Drain the engine oil

Remove the mass airflow meter

Remove the air filter housing

Remove the ignition coil cover

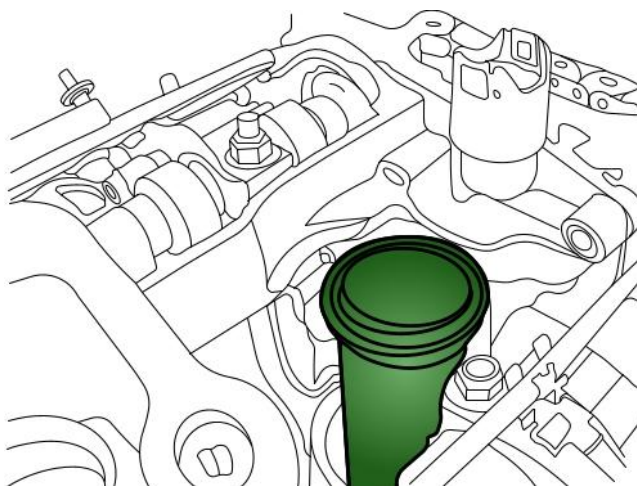
Remove the ignition coils

Remove the spark plugs

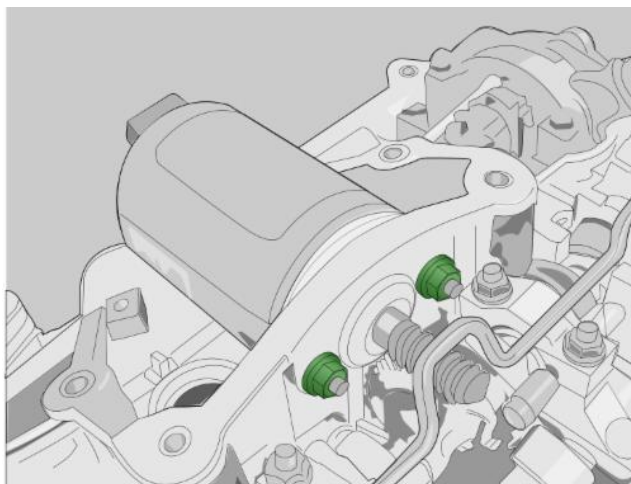
Disconnect the breather hose from the camshaft cover

Remove the cylinder head cover

Remove the spark plug housing from cylinders 1 and 4



Remove the eccentric shaft actuator

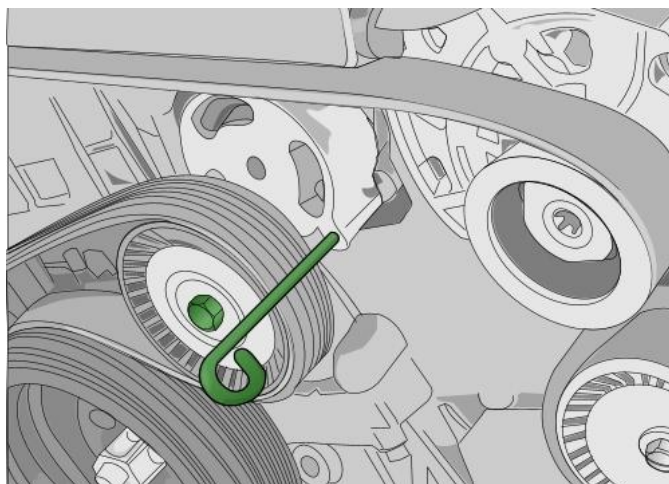


Turn the tensioner body wheel to release the belt tension

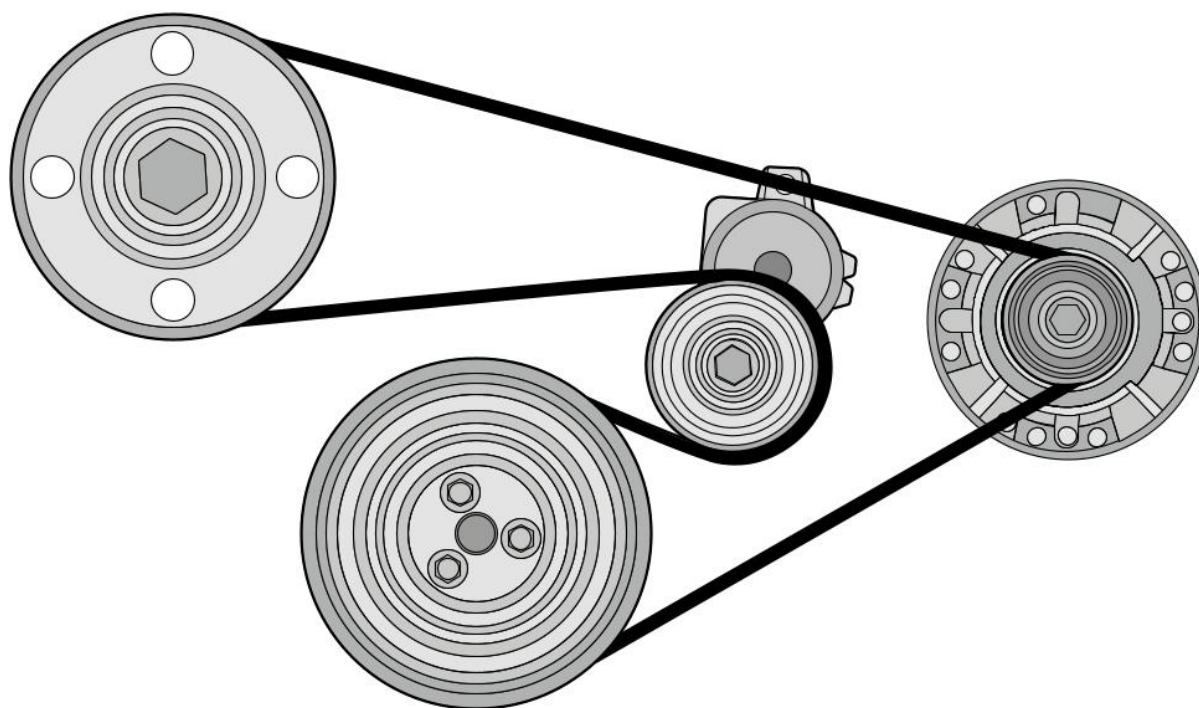
Lock the tensioner; use a locking pin (11 3 340)

Remove the ancillary drive belt

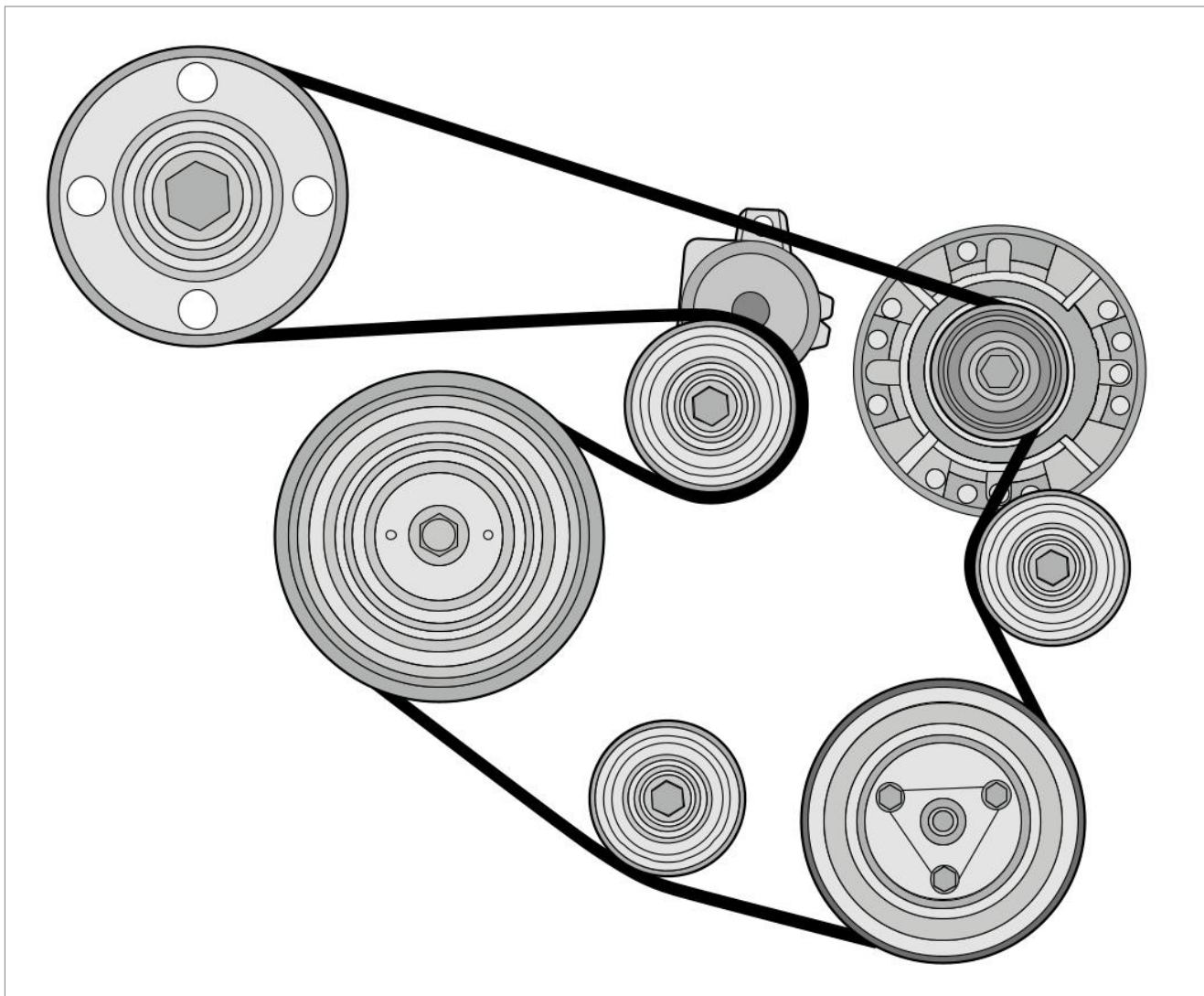
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Drive belt layout (Without air conditioning)

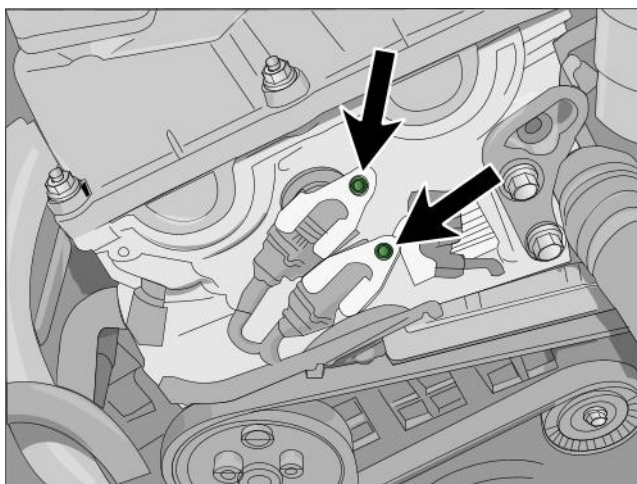


Drive belt layout (With air conditioning)

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Remove the crankshaft pulley

Remove the variable camshaft timing solenoids



Support the engine with a jack stand

Disconnect the engine supports, left and right

Raise the engine slightly (10 mm)

Remove the engine lower covers

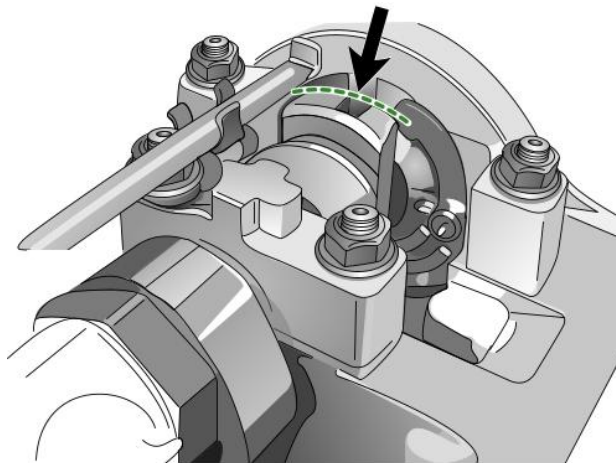
Disconnect the lower steering shaft

Support the subframe

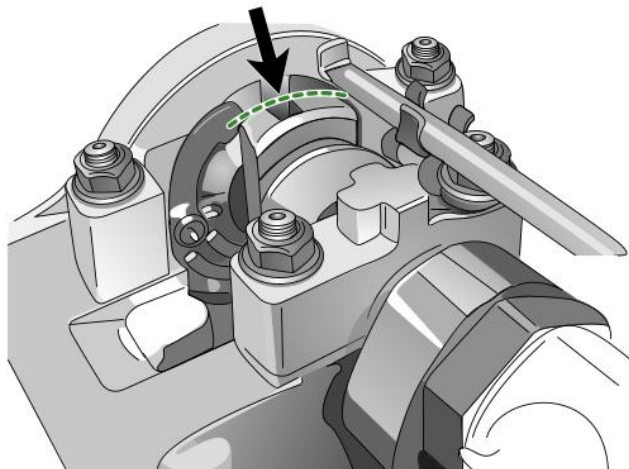
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- Fit a lifting jack
- Remove the subframe bolts
- Lower the subframe (< 10 cm)
- Remove the dipstick tube
- Remove the gearbox lower cover
- Remove the ancillary drive belt idler pulley
- Remove the oil sump
- Turn the crankshaft to TDC for cylinder 1
- Check the position of the camshafts

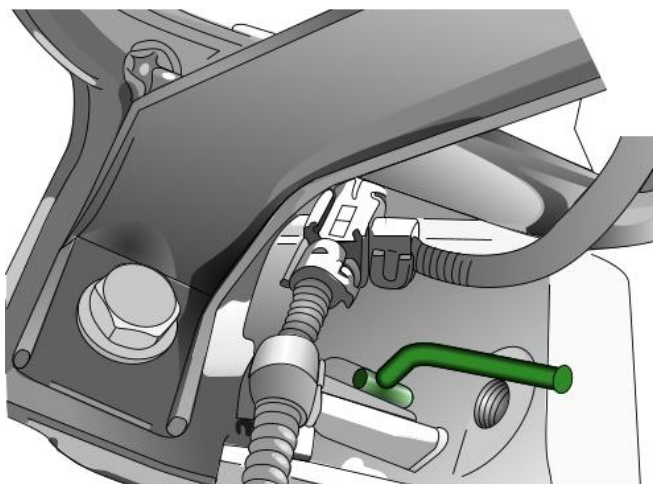
Inlet:



Exhaust:



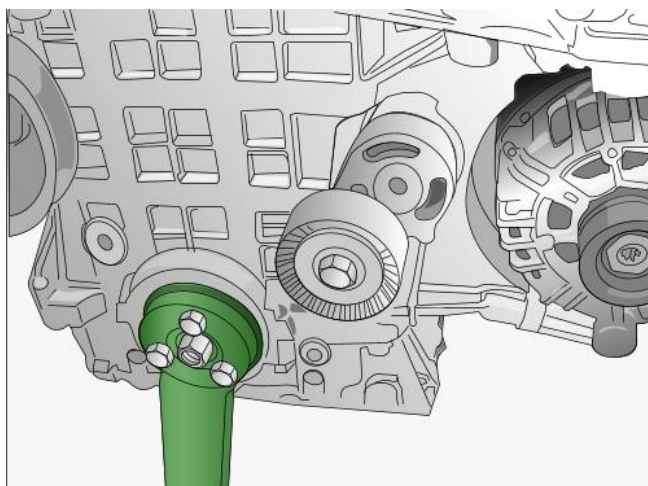
- Lock the flywheel
- Use the special tool (11 5 120)

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Fit the special tool: (11 9 280)

Remove the flywheel locking tool (11 5 120)

Loosen the centre bolt of the crankshaft gearwheel/vibration damper (90°)



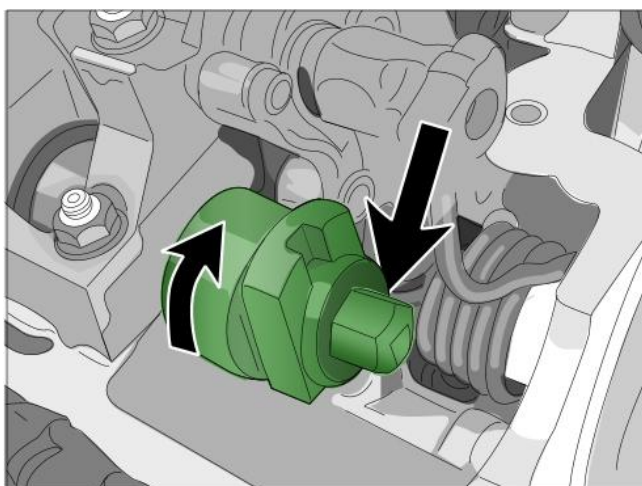
Remove the special tool (11 9 280)

Turn the crankshaft to TDC for cylinder 1

Fit the flywheel locking tool (11 5 120)

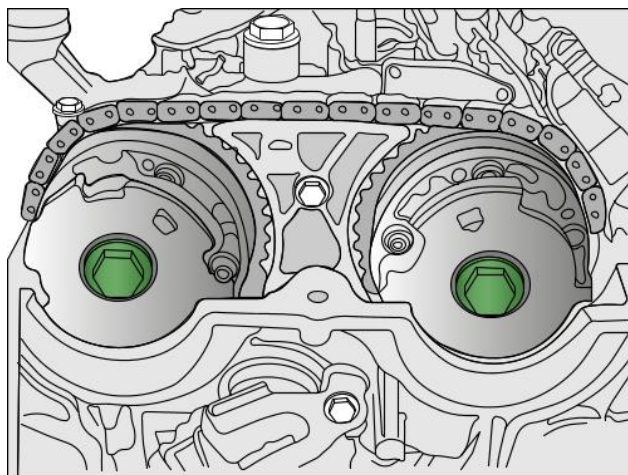
Check that the camshafts are locked

Use the special tool (11 9 270)

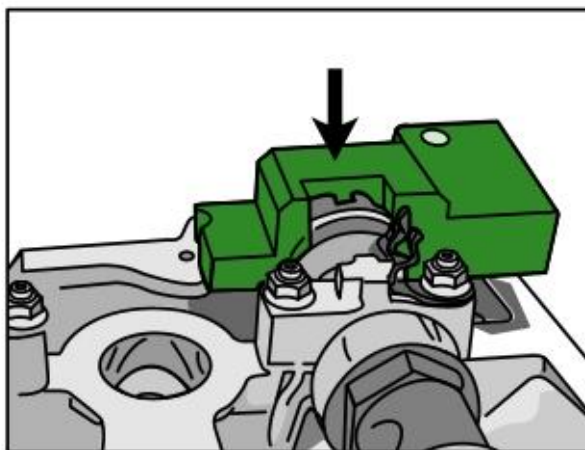


Loosen the camshaft gearwheel bolts

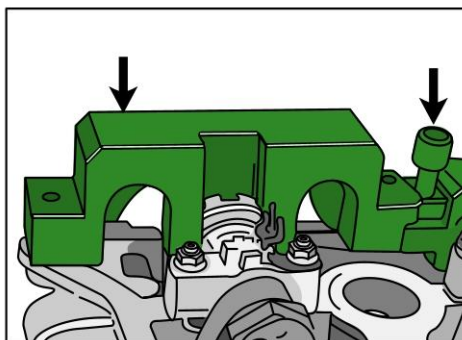
Repair Manuals



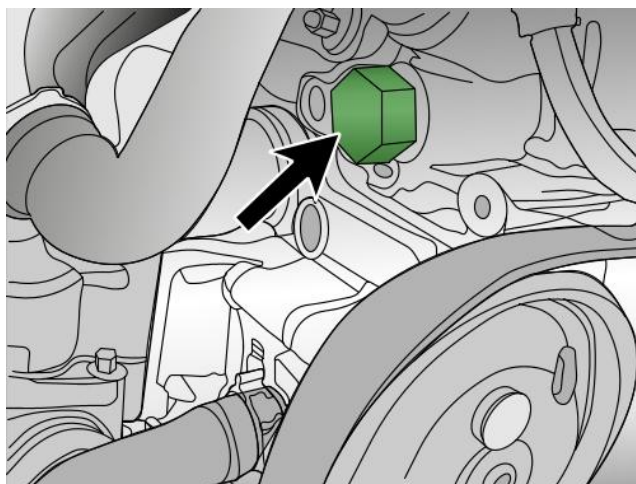
Fit the camshaft locking tools (11 9 292)



(11 9 291, 11 9 293)



Remove the chain tensioner piston

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Remove the camshaft gearwheel bolts

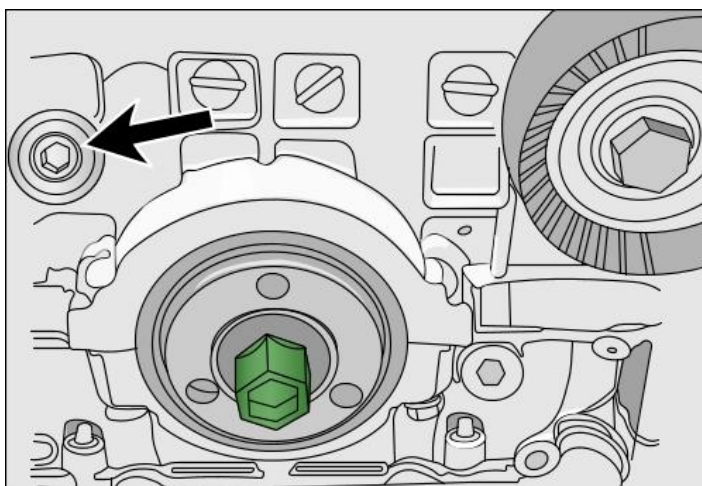
Remove the camshaft adjuster

on the left- and right-hand sides

Remove the crankshaft gearwheel centre bolt

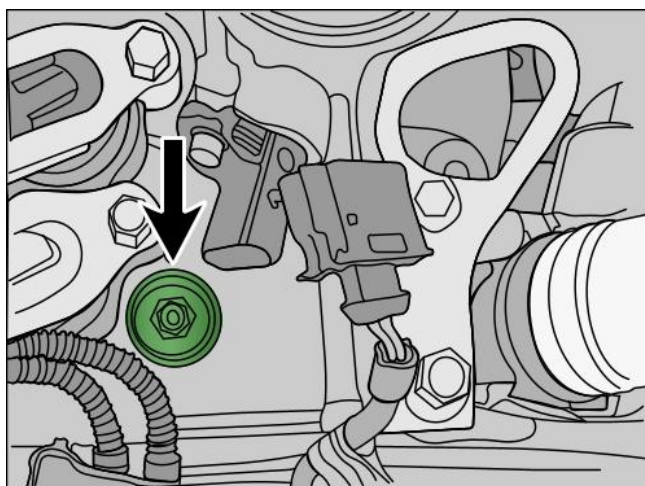
Remove the crankshaft hub

Remove the access plug



Remove the locking pin

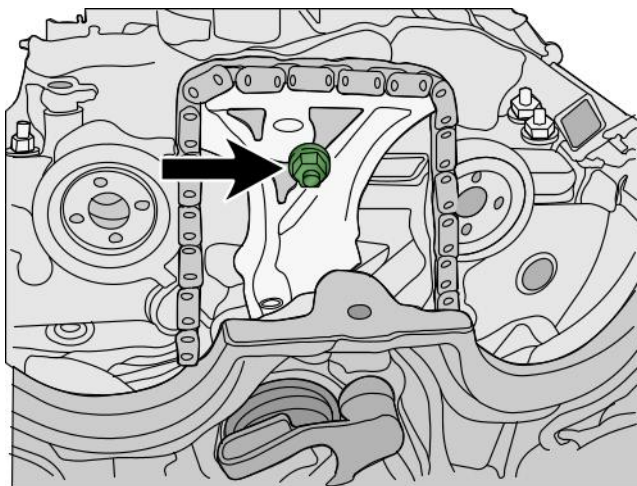
Remove the access plug



Remove the timing chain guide bolts from the cylinder head

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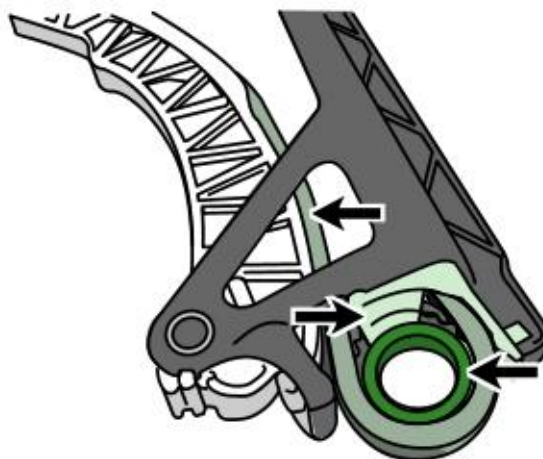
Remove the chain guide



Remove the timing chain and one of the gearwheels together

Installation

Fit the timing chain



Refit the access plug (45 Nm)

Refit the crankshaft hub

Check the condition and renew if necessary

Fit the centre bolt of the crankshaft gearwheel/vibration damper

Renew the bolt

Do not tighten!

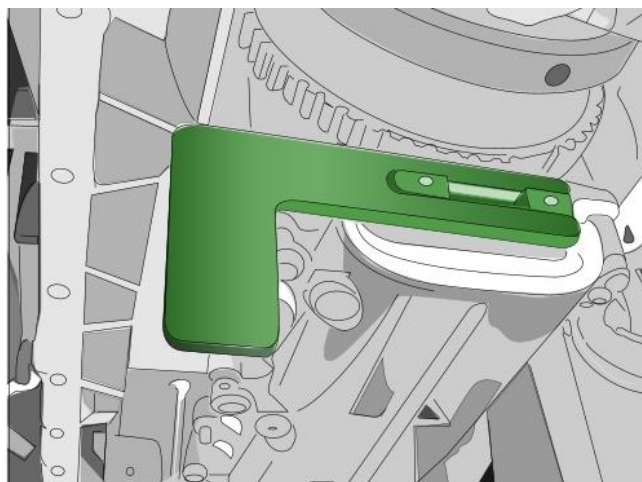
Align the balance shafts

Lock both balance shafts

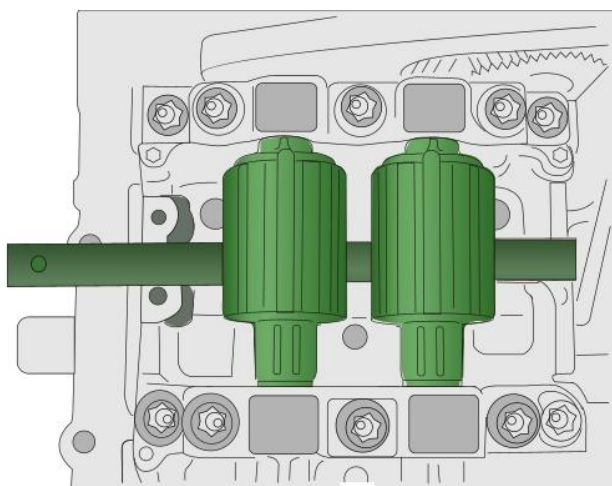
Use the special tool

Older type: (11 9 330)

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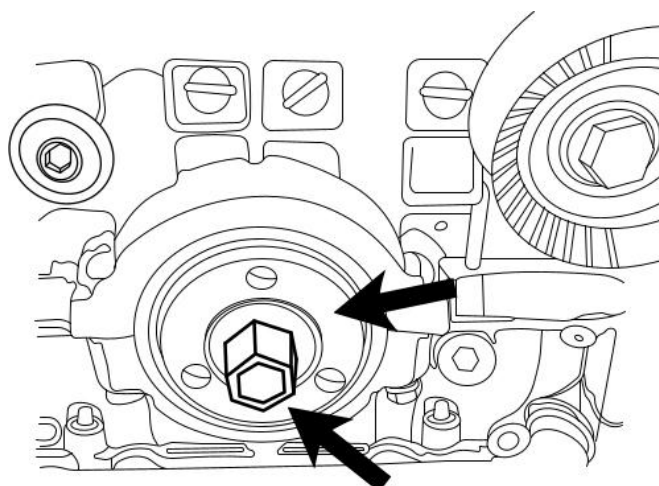


Newer type: (11 5 140)



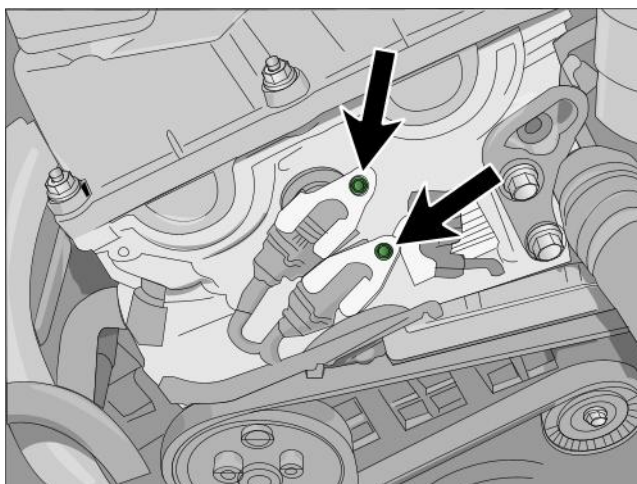
Tighten the bolt

Stage 1: (50 Nm)

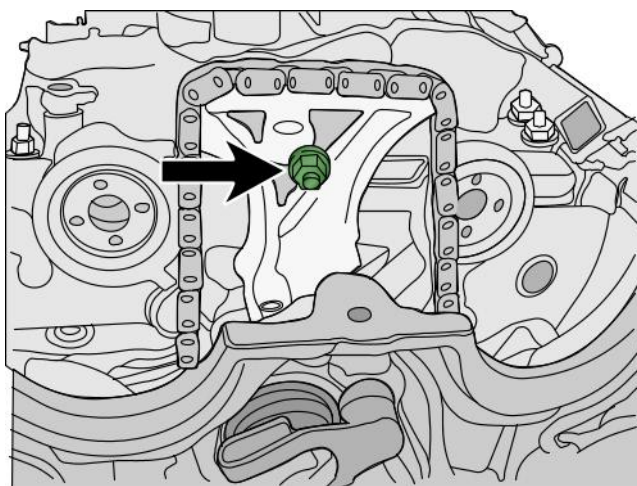


Use the special tools (11 9 280)

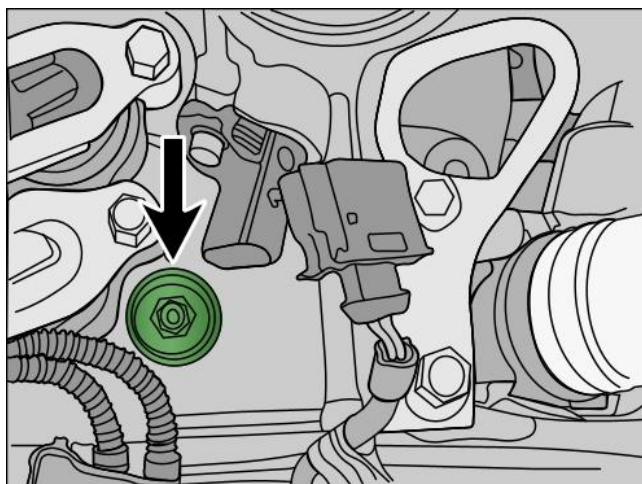
Reconnect the connector to the solenoid valve

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Refit the chain guide



Refit the access plug (25 Nm)



Refit the chain tensioner piston

Refit the inlet camshaft timing adjuster

Fit the timing chain onto the inlet camshaft gearwheel

Renew the bolt and tighten

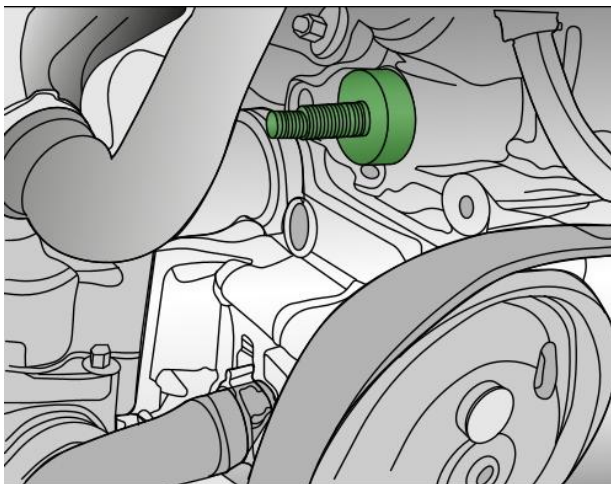
Refit the exhaust camshaft adjuster

Fit the timing chain onto the exhaust camshaft gearwheel

Renew the bolt and tighten

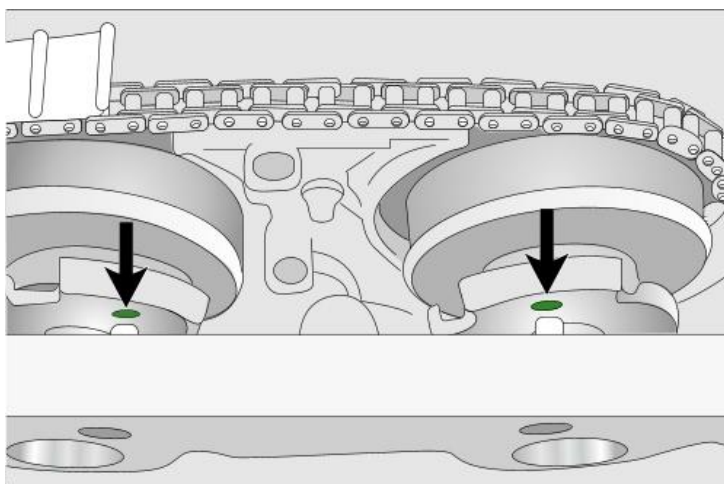
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Fit the adjustment tool (11 9 340)



Align the camshaft gearwheel marks as shown

Use the special tool (11 9 350)

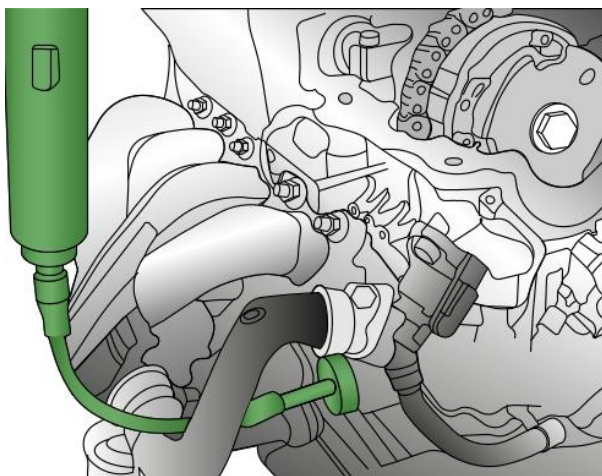


Tighten the special tool (11 9 350)

Loosen the camshaft gearwheel bolts (180°)

Pre-tension the timing chain

Use the special tool (0.6 Nm) (00 9 250)



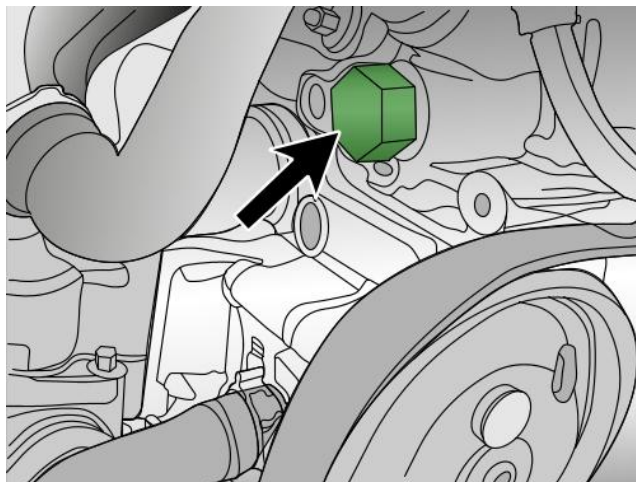
Tighten the camshaft gearwheel bolts

Remove the special tools (11 9 340, 11 9 350)

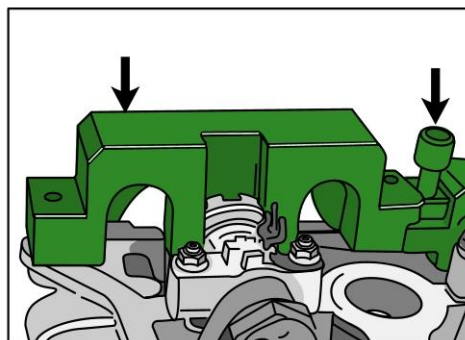
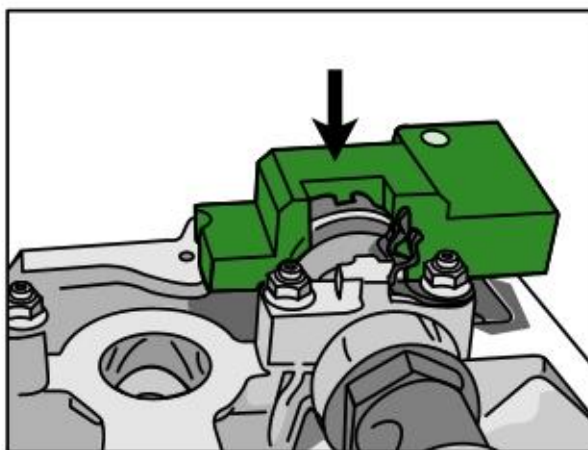
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Compress the chain tensioner piston twice to drain the oil

Refit the chain tensioner piston (65 Nm)



Remove the camshaft locking tools (11 9 292, 11 9 291, 11 9 293)



Remove the flywheel locking tool (11 5 120)

Turn the crankshaft clockwise 2 rotations

Turn the crankshaft to TDC for cylinder 1

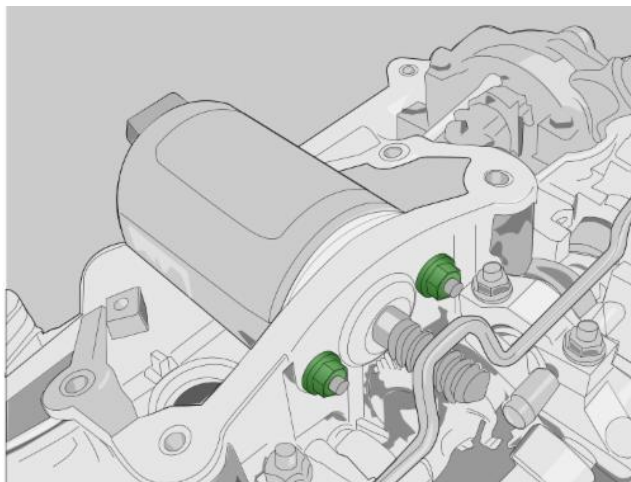
Lock the flywheel (11 5 120)

Check the timing

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The timing is set when the special tools fit correctly (11 9 292, 11 9 291, 11 9 293)

Refit the eccentric shaft actuator



Refit the cylinder head cover

Refit the spark plugs

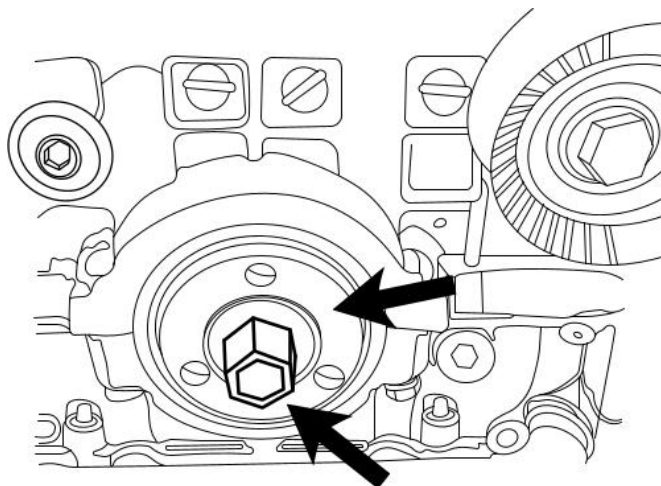
Refit the ignition coils

Refit the ignition coil covers

Remove the flywheel locking tool (11 5 120)

Tighten the bolt (150°)

Use the special tool (11 9 280)



Refit the oil sump

Renew the bolts

Fit the ancillary drive belt pulley

Refit the gearbox lower cover

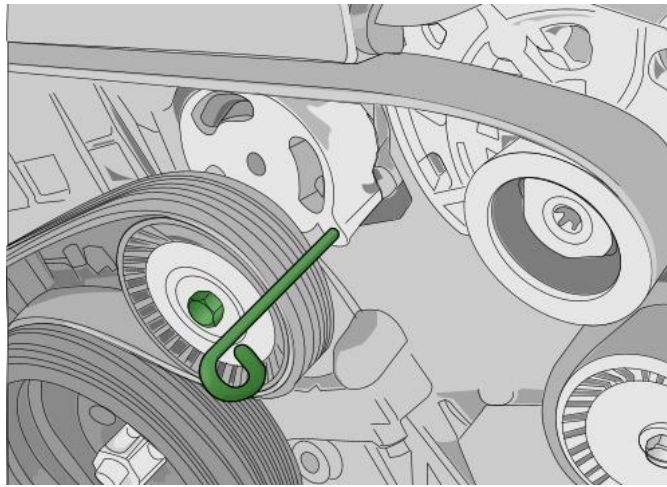
Refit the dipstick tube

Fit the vibration damper (34 Nm)

Refit the ancillary drive belt

Remove the tensioner locking tool (11 3 340)

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Refit the air filter housing

Refit the mass airflow meter

Refill the engine oil

Engine sump, including filter 4.25 (l)

Reconnect the battery earth cable

Torque settings

Camshaft gearwheel

Camshaft gearwheel

Renew the bolts

Stage 1 20 (Nm)

Stage 2 180 (°)

Crankshaft gearwheel

Crankshaft gearwheel (Centre bolt)

Renew the bolt

Stage 1 50 (Nm)

Stage 2 150 (°)

Crankshaft pulley

Crankshaft pulley (Outer bolts) 35 (Nm)

Oil sump

Sump 30 (Nm)

Timing chain tensioner

Timing chain tensioner (Renew the seal) 65 (Nm)

Timing chain guide

Timing chain guide (Lower) 20 (Nm)

Timing chain guide (Upper) 14 (Nm)

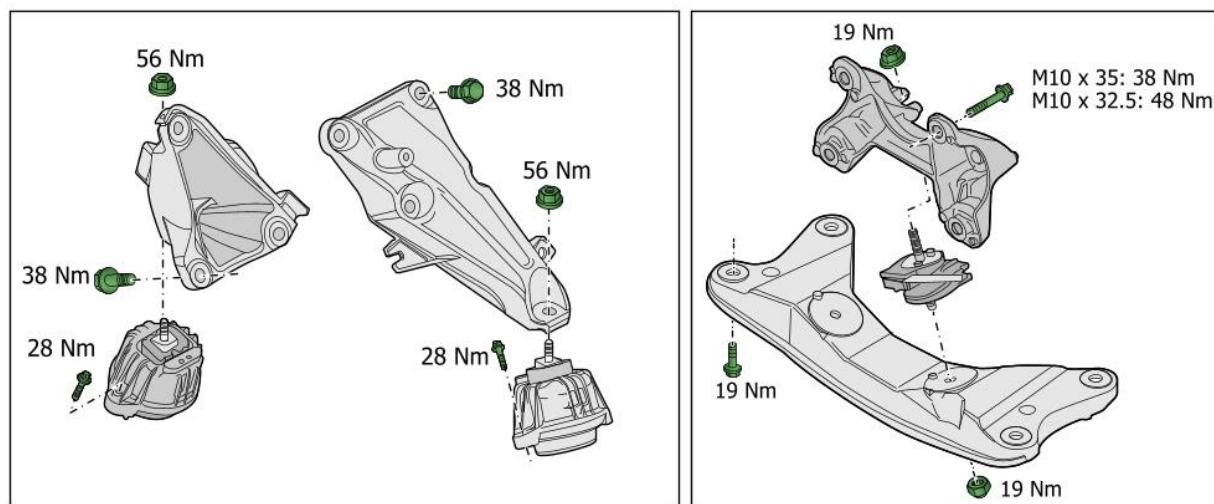
Steering: Refer to 'Technical drawings'

Suspension: Refer to 'Technical drawings'

Engine/gearbox mounts

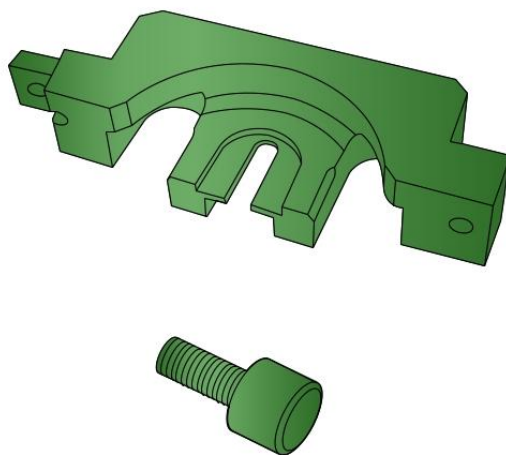
Engine

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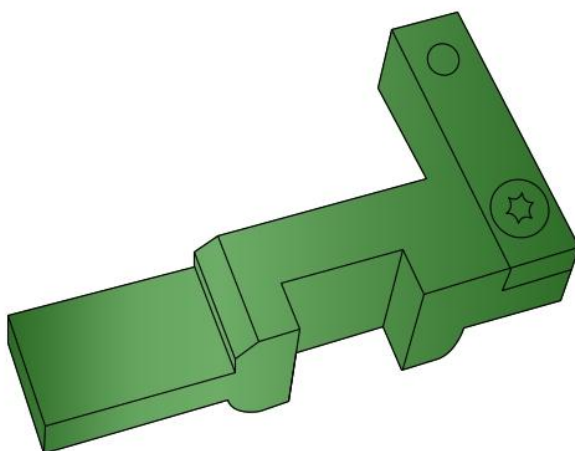


Special tools

Camshaft locking tool: (11 9 291, 11 9 293)

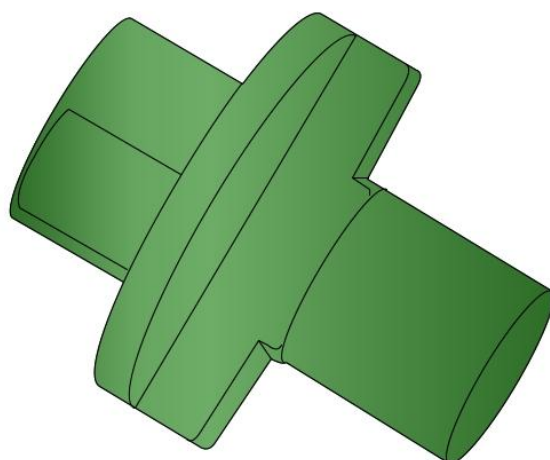


Camshaft locking tool: (11 9 292)

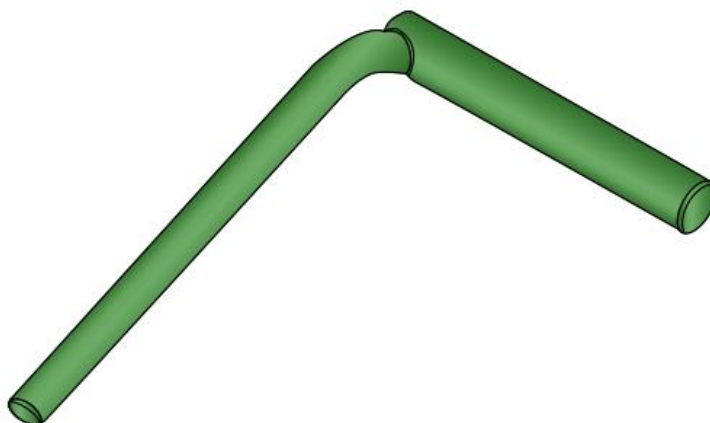


Camshaft turning tool: (11 9 270)

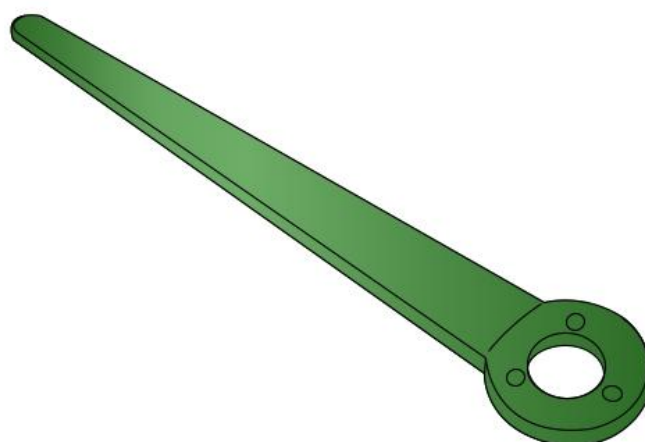
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Flywheel locking tool: (11 5 120)

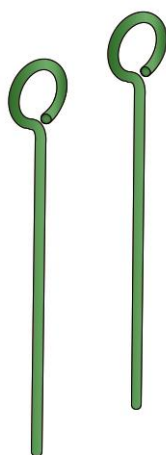


Crankshaft centering tool: (11 9 280)



Chain tensioner locking pin: (11 3 340)

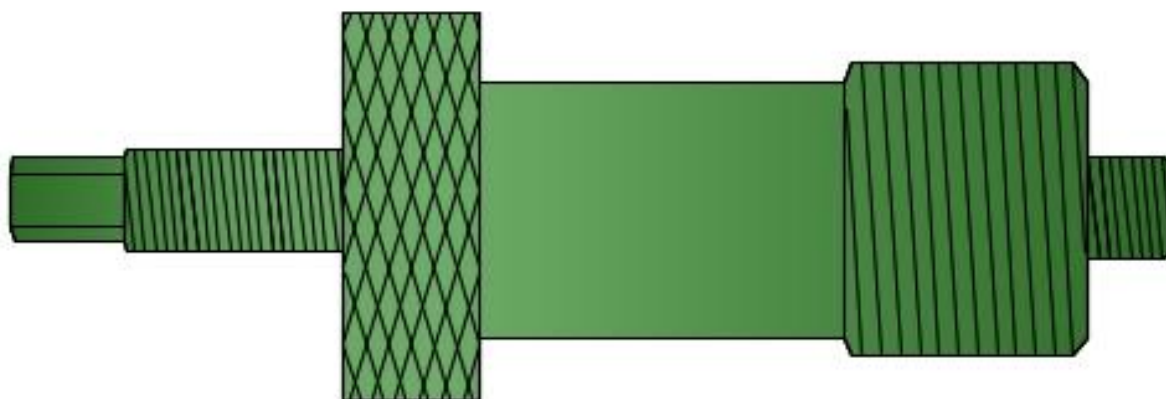
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Balance shaft tensioning tool: (11 5 140)

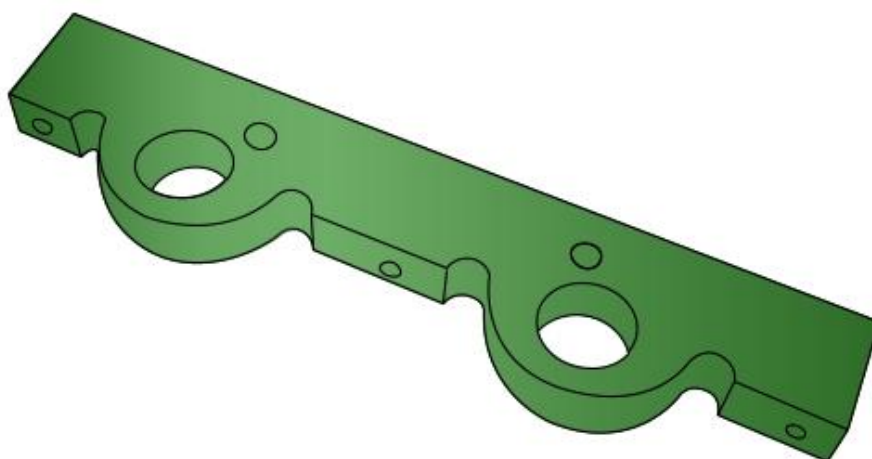


Adaptor: (11 9 340)

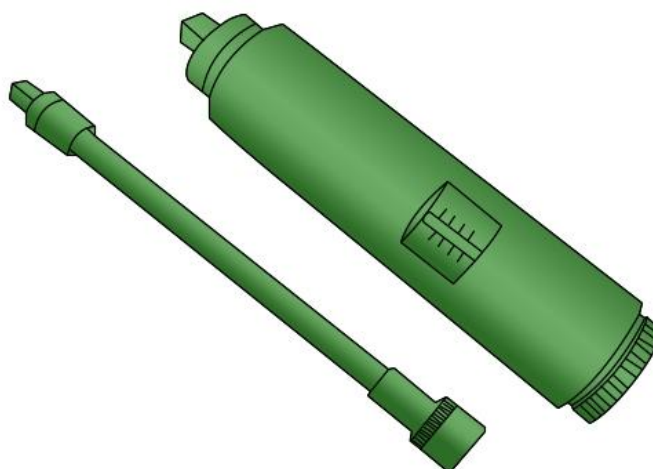


Camshaft alignment tool: (11 9 350)

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Adjustment tool: (00 9 250)



Balance shaft tensioning tool: (11 9 330)

